PYTHON PROGRAMMING

ACTIVITY-BASED LEARNING (ABL) ASSIGNMENT

DR. RVR NRI INSTITUTE OF TECHNOLOGY

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Student Pass/Fail Checker

Python file: a1.py **Concept:** if-else

Problem Statement

Check whether a student passes based on the marks entered.

Objective

Display Pass if marks are 35 or above; otherwise display Fail.

Algorithm

1. Input marks.

2. If marks >= 35, display Pass; otherwise, display Fail.

Pseudocode

```
START
INPUT marks
IF marks >= 35
    DISPLAY "Pass"
ELSE
    DISPLAY "Fail"
END IF
STOP
```

Python Source Code

```
marks = int(input("Enter marks: "))

if marks >= 35:
    print("Pass")
else:
    print("Fail")
```

Sample Input

```
Enter marks: 75
```

Expected Output — Text Only

```
Pass
```

Test Cases

Input / Test	Expected Result
75	Pass
28	Fail

Conclusion

This case demonstrates the stated Python concept.

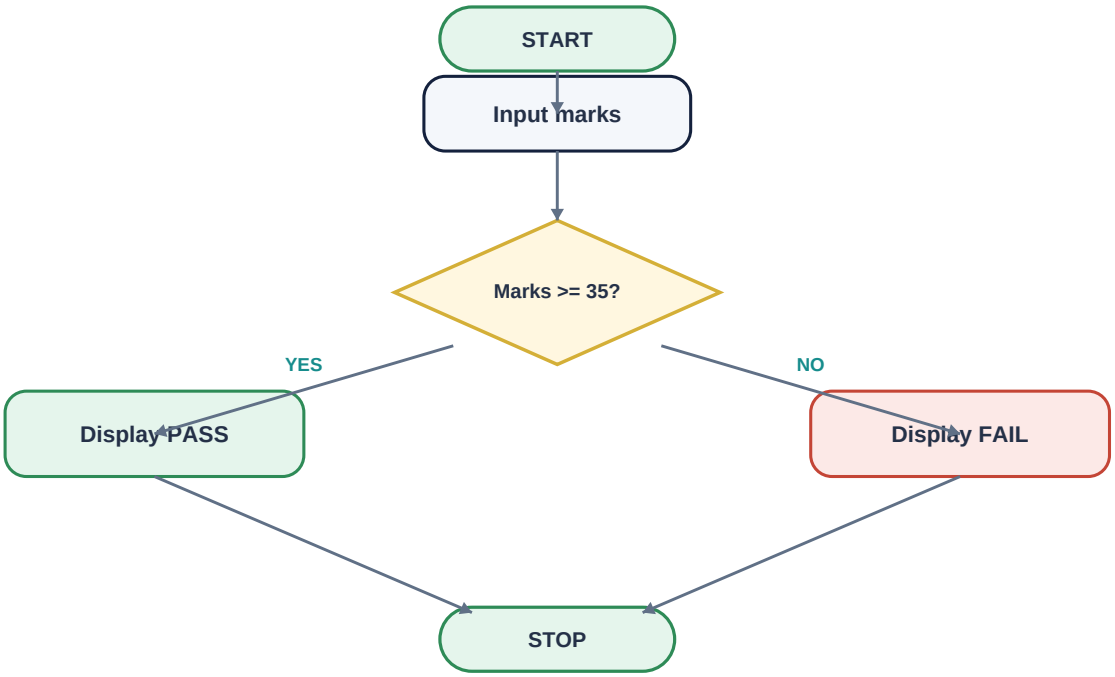
Experiment 1: Student Pass/Fail Checker

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 1 | FLOWCHART

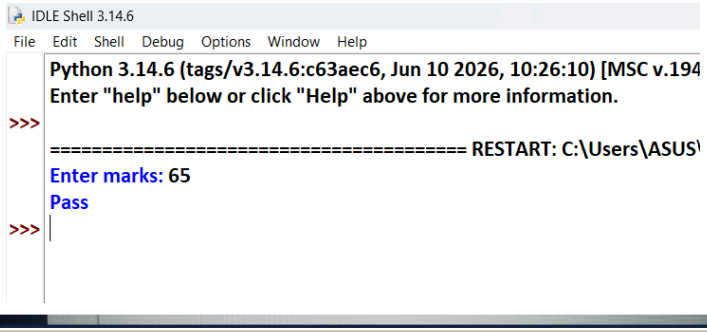
Student Pass/Fail Checker



ASSUMPTIONS

Marks are entered as a whole number from 0 to 100. A score of 35 or more is a pass.

PYTHON IDLE OUTPUT SCREENSHOT



Exam Eligibility Checker

Python file: a2.py **Concept:** if-else

Problem Statement

Check exam eligibility using the attendance percentage.

Objective

Display Eligible for exams when attendance is 75% or above.

Algorithm

1. Input attendance percentage.

2. If attendance >= 75, display Eligible for exams; otherwise, display Not eligible for exams.

Pseudocode

```
START
INPUT attendance
IF attendance >= 75
    DISPLAY "Eligible for exams"
ELSE
    DISPLAY "Not eligible for exams"
END IF
STOP
```

Python Source Code

```
attendance = int(input("Enter attendance percentage: "))

if attendance >= 75:
    print("Eligible for exams")
else:
    print("Not eligible for exams")
```

Sample Input

```
Enter attendance percentage: 80
```

Expected Output — Text Only

```
Eligible for exams
```

Test Cases

Input / Test	Expected Result
80	Eligible for exams
65	Not eligible for exams

Conclusion

This case demonstrates the stated Python concept.

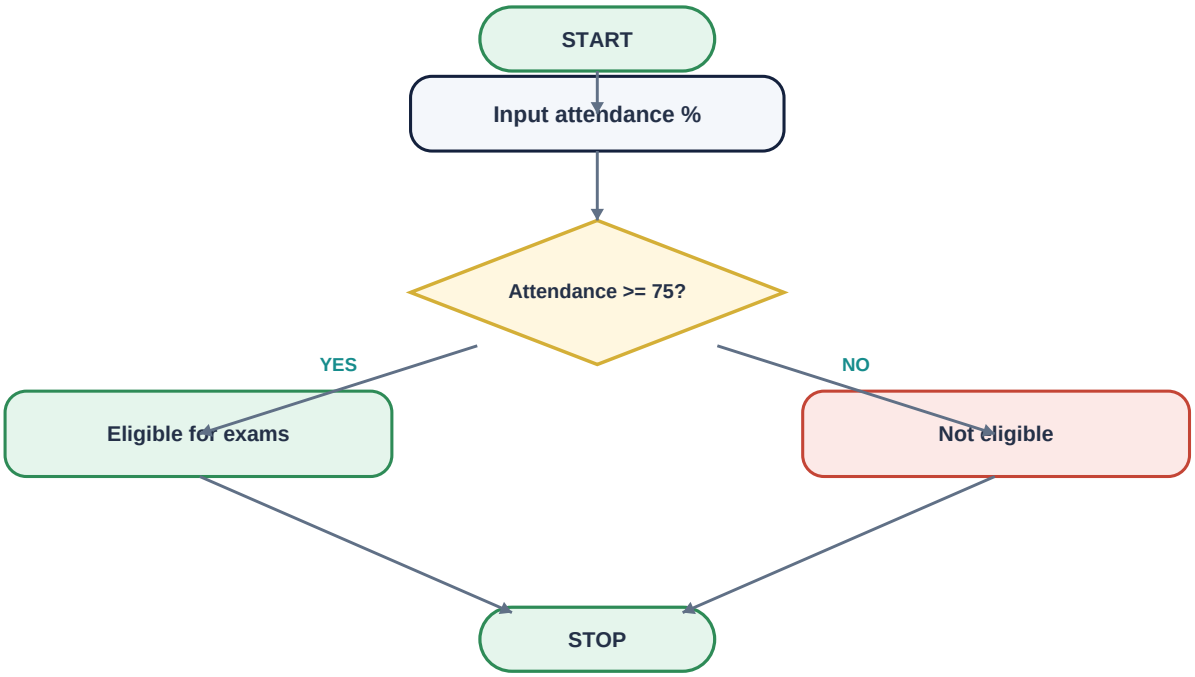
Experiment 2: Exam Eligibility Checker

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 2 | FLOWCHART

Exam Eligibility Checker



ASSUMPTIONS

Attendance is entered as a percentage from 0 to 100. The eligibility threshold is 75%.

PYTHON IDLE OUTPUT SCREENSHOT

```
Enter attendance percentage: 98
Eligible for exams
>>
```

Scholarship Eligibility Calculator

Python file: a3.py **Concept:** if-elif-else, logical operator and

Problem Statement

Determine scholarship eligibility using marks and family income.

Objective

Apply the conditions in the program to display full, half, or no scholarship.

Algorithm

1. Input marks and family income.

2. If marks >= 90 and income <= 200000, display Full scholarship.

3. Otherwise, if marks >= 75 and income <= 300000, display Half scholarship.

4. Otherwise, display No scholarship.

Pseudocode

```
START
INPUT marks, income
IF marks >= 90 AND income <= 200000
    DISPLAY "Full scholarship"
ELSE IF marks >= 75 AND income <= 300000
    DISPLAY "Half scholarship"
ELSE
    DISPLAY "No scholarship"
END IF
STOP
```

Python Source Code

```
marks = int(input("Enter marks: "))
income = int(input("Enter family income: "))

if marks >= 90 and income <= 200000:
    print("Full scholarship")
elif marks >= 75 and income <= 300000:
    print("Half scholarship")
else:
    print("No scholarship")
```

Sample Input

```
Enter marks: 95
Enter family income: 150000
```

Expected Output — Text Only

```
Full scholarship
```

Test Cases

Input / Test	Expected Result
95, 150000	Full scholarship
80, 250000	Half scholarship
70, 400000	No scholarship

Conclusion

This case demonstrates the stated Python concept.

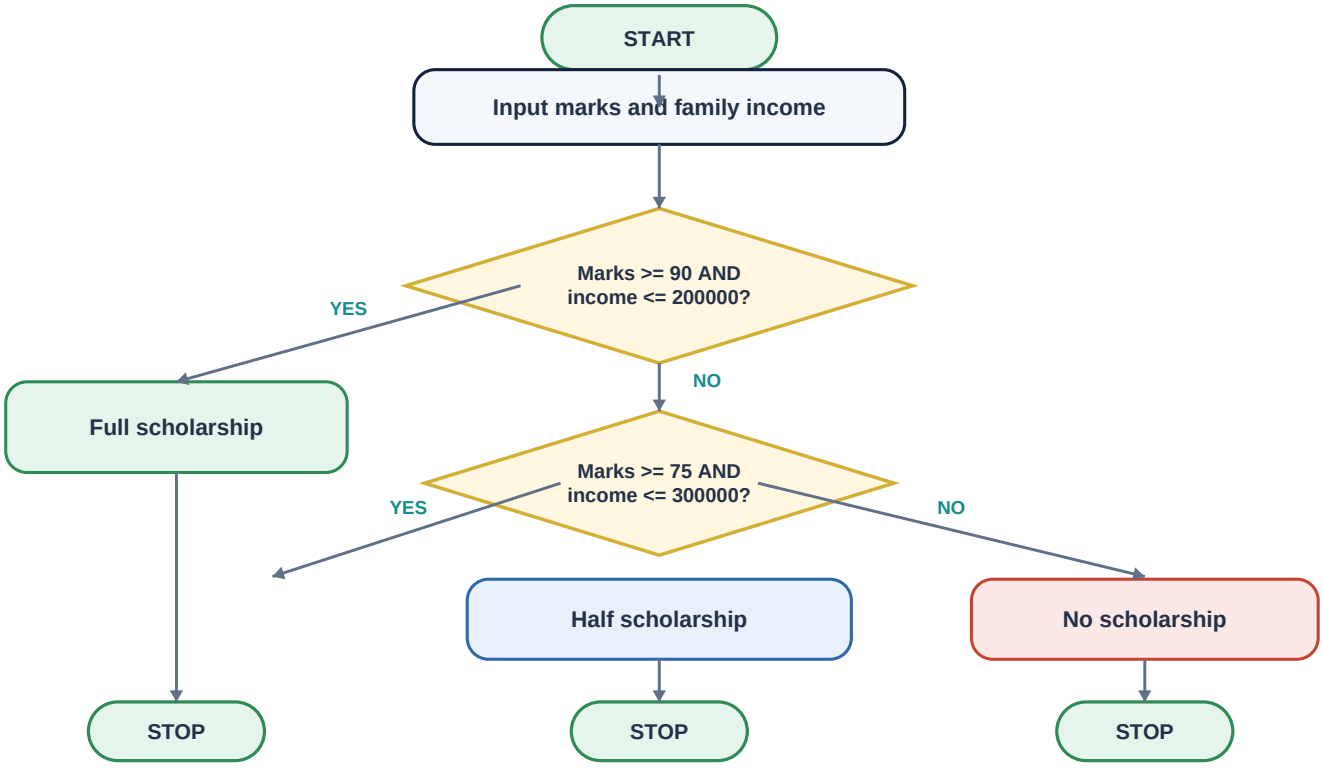
Experiment 3: Scholarship Eligibility Calculator

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 3 | FLOWCHART

Scholarship Eligibility Calculator



ASSUMPTIONS

Marks and family income are valid non-negative numbers. The higher scholarship condition is checked first.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\A
Enter marks: 98
Enter family income: 100000
Full scholarship
>
```

Electricity Bill Calculator

Python file: a4.py **Concept:** if-elif-else

Problem Statement

Calculate an electricity bill based on units consumed.

Objective

Use the rates specified in the supplied code: up to 100 units at 2 per unit, up to 200 units at 3 per unit, and above 200 units at 5 per unit.

Algorithm

1. Input units consumed.
2. If units <= 100, bill = units * 2.
3. Otherwise, if units <= 200, bill = units * 3.
4. Otherwise, bill = units * 5.
5. Display the bill.

Pseudocode

```
START
INPUT units
IF units <= 100
    bill = units * 2
ELSE IF units <= 200
    bill = units * 3
ELSE
    bill = units * 5
END IF
DISPLAY bill
STOP
```

Python Source Code

```
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 2
elif units <= 200:
    bill = units * 3
else:
    bill = units * 5

print("Electricity bill =", bill)
```

Sample Input

```
Enter units consumed: 80
```

Expected Output — Text Only

```
Electricity bill = 160
```

Test Cases

Input / Test	Expected Result
80	Electricity bill = 160
150	Electricity bill = 450
250	Electricity bill = 1250

Conclusion

This case demonstrates the stated Python concept.

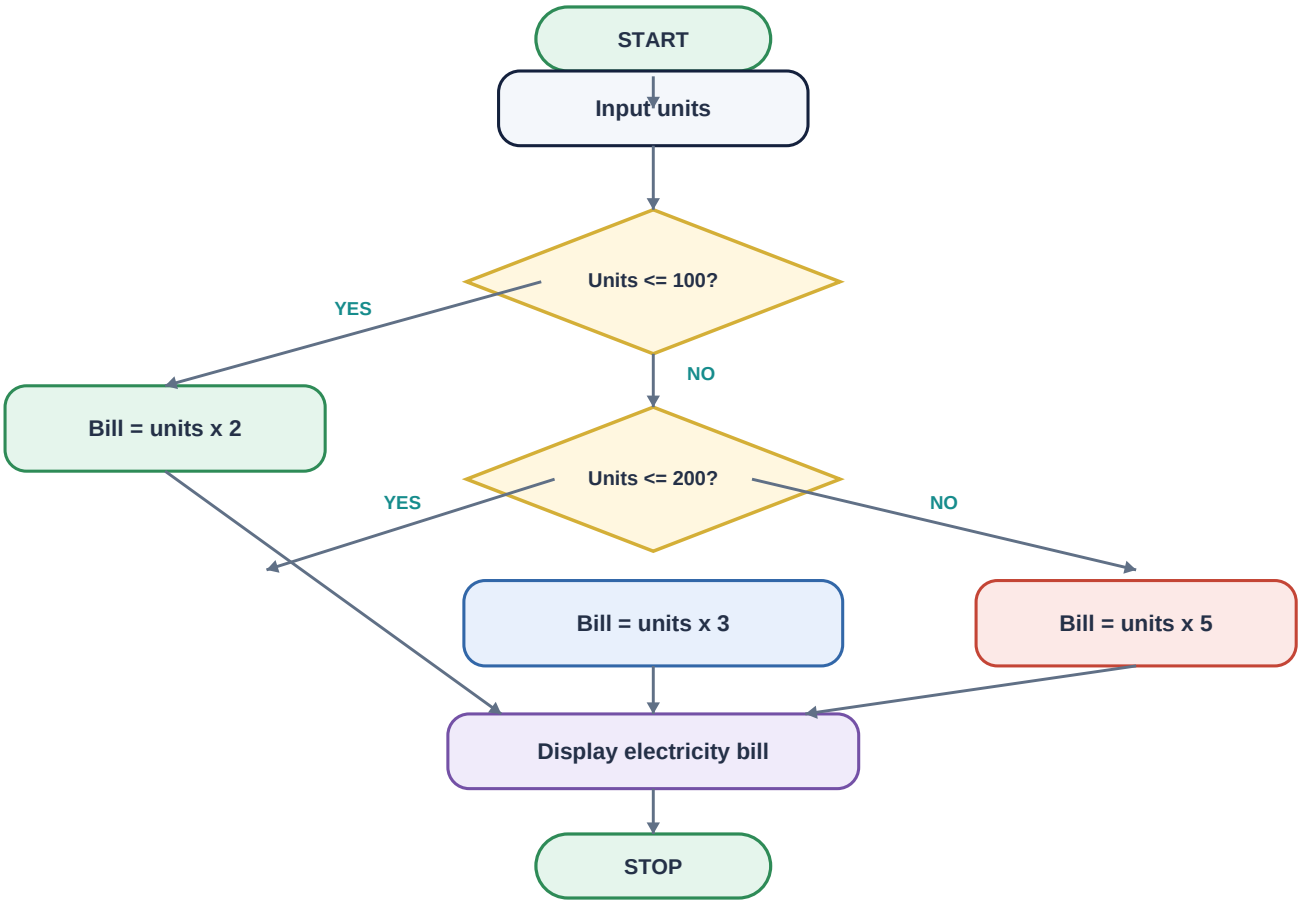
Experiment 4: Electricity Bill Calculator

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 4 | FLOWCHART

Electricity Bill Calculator



ASSUMPTIONS

Units consumed are non-negative whole numbers. The program applies one rate to all units based on the selected range.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\Ap
Enter units consumed: 123
Electricity bill = 369
>>>
```

Hospital Bill Discount Calculator

Python file: a5.py **Concept:** if-else, arithmetic

Problem Statement

Calculate a discount and final amount for a hospital bill.

Objective

Apply a 10% discount when the bill is ■5,000 or more; otherwise apply no discount.

Algorithm

1. Input hospital bill.

2. If bill >= 5000, calculate a 10% discount; otherwise set discount to 0.

3. Subtract the discount from the bill.

4. Display discount and final bill.

Pseudocode

```
START
INPUT bill
IF bill >= 5000
    discount = bill * 10 / 100
ELSE
    discount = 0
END IF
total = bill - discount
DISPLAY discount, total
STOP
```

Python Source Code

```
bill = int(input("Enter hospital bill: "))

if bill >= 5000:
    discount = bill * 10 / 100
else:
    discount = 0

total = bill - discount

print("Discount =", discount)
print("Final bill =", total)
```

Sample Input

```
Enter hospital bill: 6000
```

Expected Output — Text Only

```
Discount = 600.0
Final bill = 5400.0
```

Test Cases

Input / Test	Expected Result
6000	Discount = 600.0; Final bill = 5400.0
4000	Discount = 0; Final bill = 4000

Conclusion

This case demonstrates the stated Python concept.

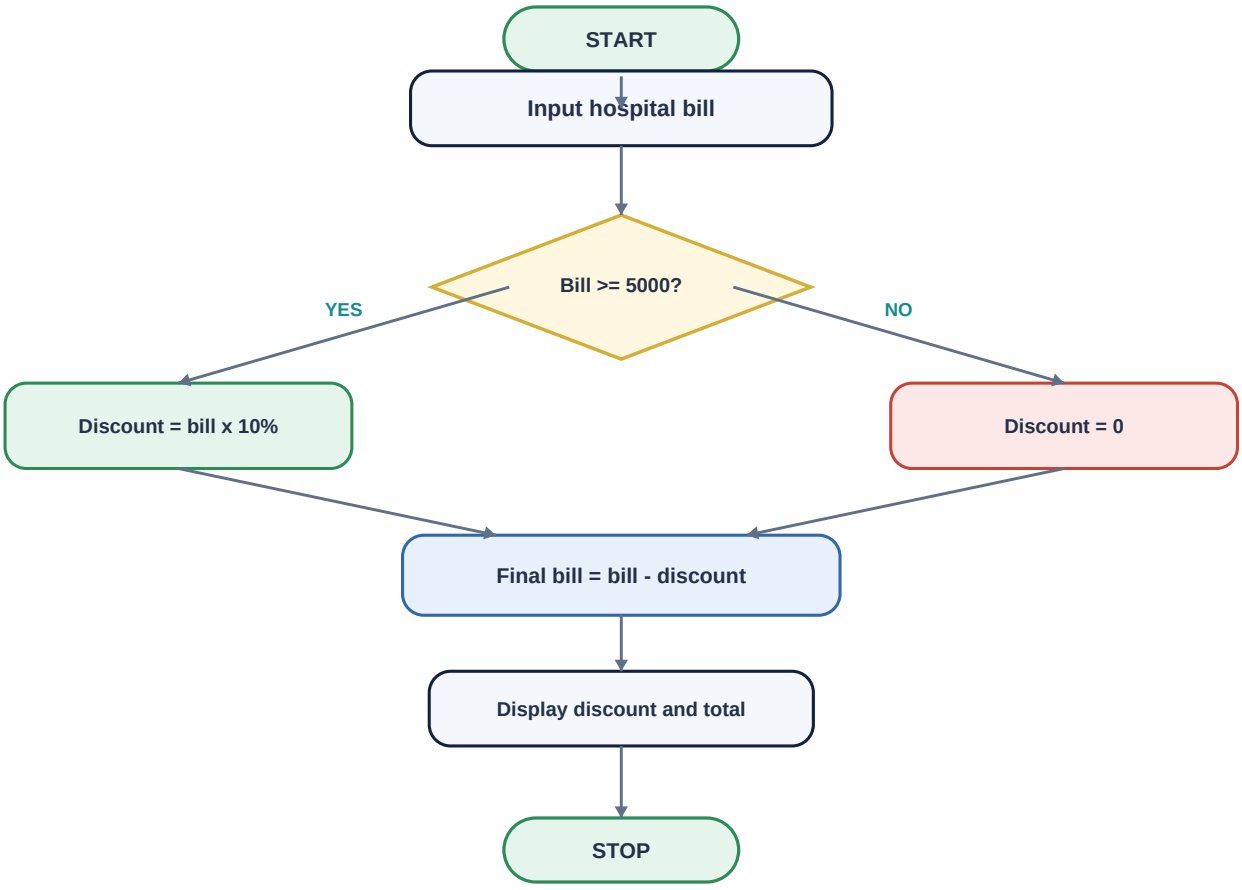
Experiment 5: Hospital Bill Discount Calculator

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 5 | FLOWCHART

Hospital Bill Discount Calculator



ASSUMPTIONS

The hospital bill is non-negative. Bills of 5,000 or more receive a 10% discount.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\App
Enter hospital bill: 12000
Discount = 1200.0
Final bill = 10800.0
>>
```

Employee Salary Allowance Calculator

Python file: a6.py **Concept:** if-else

Problem Statement

Calculate an employee's allowance and total salary from the basic salary.

Objective

Give an allowance of 5000 when salary is at least 20000; otherwise give 2000.

Algorithm

1. Input basic salary.

2. If salary >= 20000, set allowance to 5000; otherwise set it to 2000.

3. Add allowance to salary.

4. Display allowance and total salary.

Pseudocode

```
START
INPUT salary
IF salary >= 20000
    allowance = 5000
ELSE
    allowance = 2000
END IF
total = salary + allowance
DISPLAY allowance, total
STOP
```

Python Source Code

```
salary = int(input("Enter basic salary: "))

if salary >= 20000:
    allowance = 5000
else:
    allowance = 2000

total = salary + allowance

print("Allowance =", allowance)
print("Total salary =", total)
```

Sample Input

```
Enter basic salary: 25000
```

Expected Output — Text Only

```
Allowance = 5000
Total salary = 30000
```

Test Cases

Input / Test	Expected Result
25000	Allowance = 5000; Total salary = 30000
15000	Allowance = 2000; Total salary = 17000

Conclusion

This case demonstrates the stated Python concept.

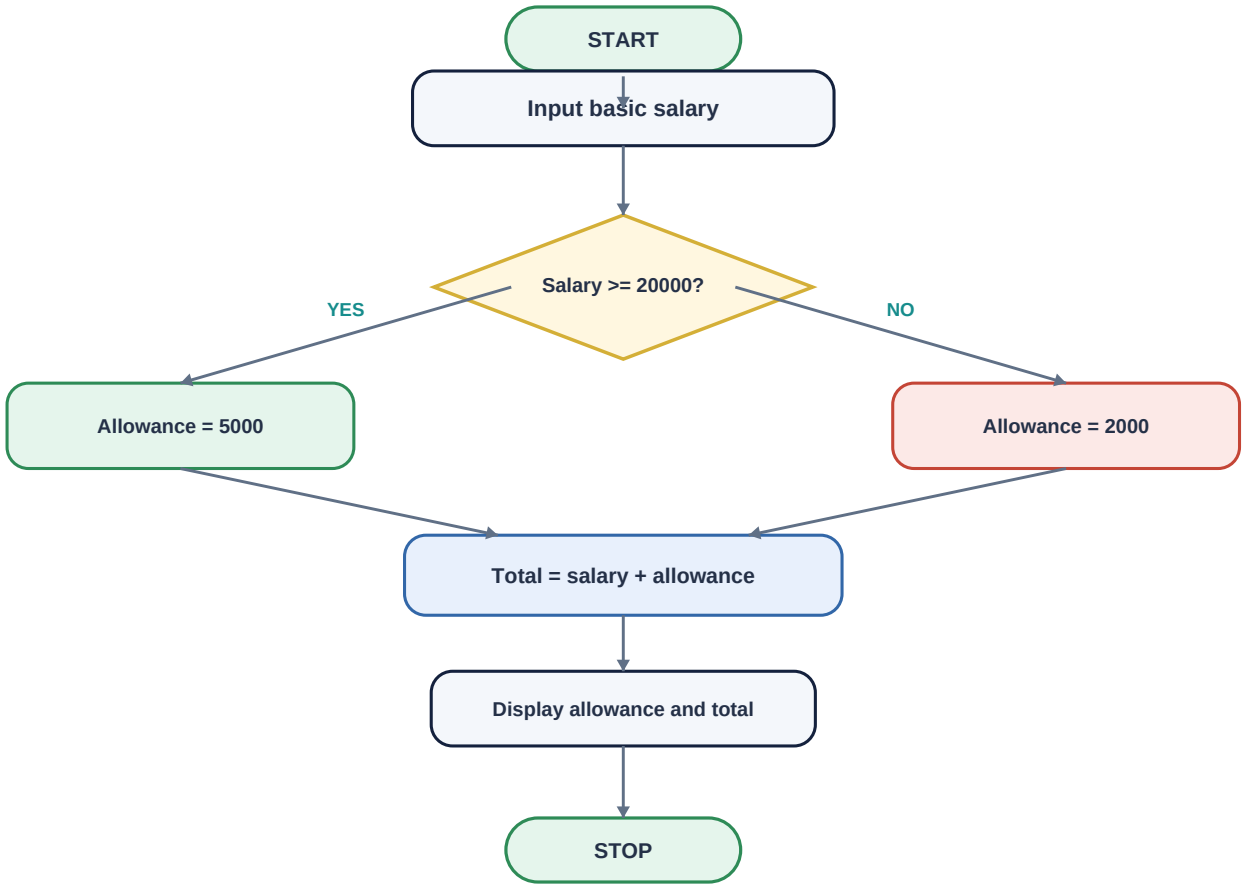
Experiment 6: Employee Salary Allowance Calculator

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART

EXPERIMENT 6 | FLOWCHART

Employee Salary Allowance Calculator



ASSUMPTIONS

Basic salary is a non-negative amount. Salary of 20,000 or more receives 5,000 allowance; otherwise 2,000.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\AppData\Local\Programs\Python\Pytho
Enter basic salary: 120000
Allowance = 5000
Total salary = 125000
>>>
```

Mini ATM System Simulation

Python file: a7.py **Concept:** if-elif-else, nested if

Problem Statement

Simulate checking balance, depositing money, and withdrawing money.

Objective

Use menu selection to update a starting balance of 5000 and reject withdrawals greater than the balance.

Algorithm

1. Set balance = 5000 and display the menu.

2. Input choice.

3. Choice 1 displays the balance; choice 2 deposits an amount; choice 3 withdraws only if sufficient funds exist.

4. For any other choice, display Invalid choice.

Pseudocode

```
START
balance = 5000
DISPLAY menu
INPUT choice
IF choice == 1
    DISPLAY balance
ELSE IF choice == 2
    INPUT amount
    balance = balance + amount
    DISPLAY balance
ELSE IF choice == 3
    INPUT amount
    IF amount <= balance
        balance = balance - amount
        DISPLAY balance
    ELSE
        DISPLAY "Insufficient balance"
    END IF
ELSE
    DISPLAY "Invalid choice"
END IF
STOP
```

Python Source Code

```
balance = 5000

print("1. Check Balance")
print("2. Deposit")
print("3. Withdraw")

choice = int(input("Enter choice: "))

if choice == 1:
    print("Balance =", balance)
elif choice == 2:
    amount = int(input("Enter deposit amount: "))
    balance = balance + amount
    print("Balance =", balance)
elif choice == 3:
    amount = int(input("Enter withdrawal amount: "))
    if amount <= balance:
        balance = balance - amount
        print("Balance =", balance)
    else:
        print("Insufficient balance")
else:
    print("Invalid choice")
```

Sample Input

```
Enter choice: 2
Enter deposit amount: 1000
```

Expected Output — Text Only

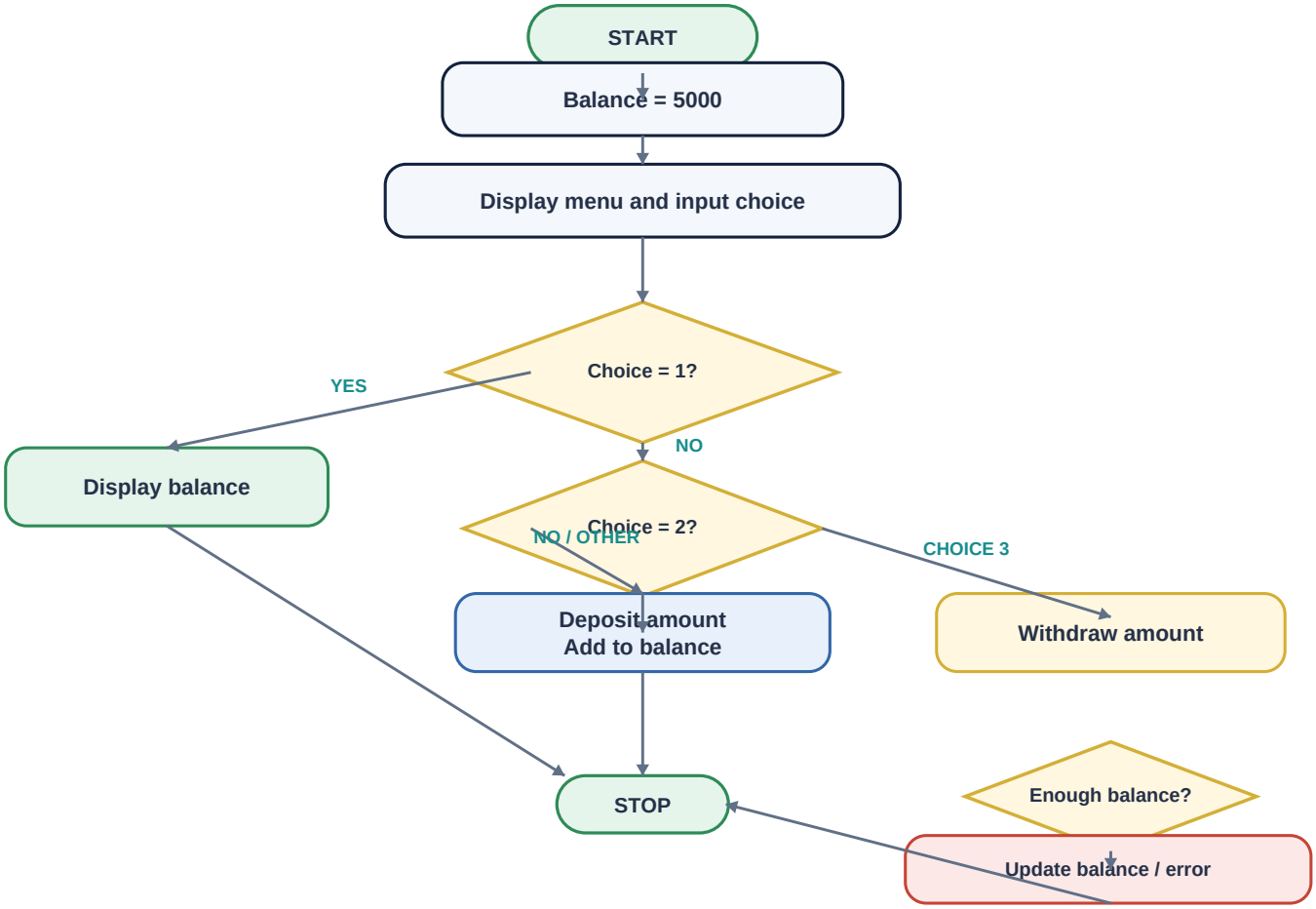
```
1. Check Balance
```

```
2. Deposit
3. Withdraw
Balance = 6000
```

Test Cases

EXPERIMENT 7 | FLOWCHART

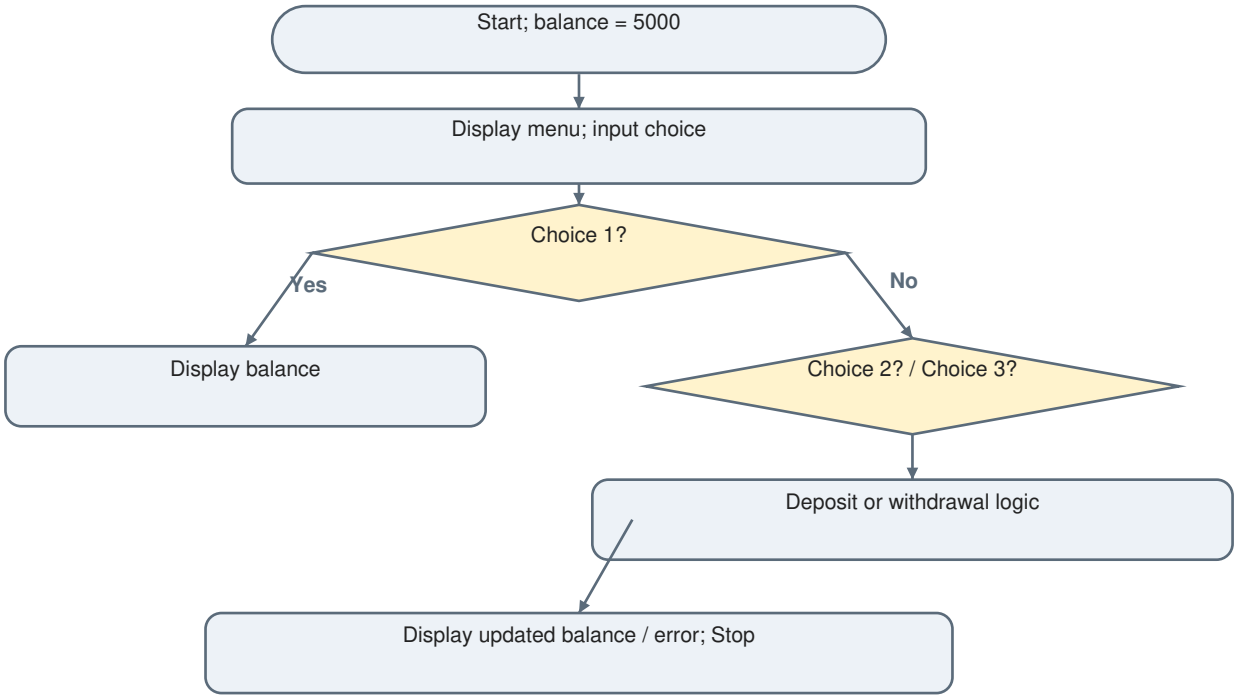
Mini ATM System Simulation



Experiment 7: Mini ATM System Simulation

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART



ASSUMPTIONS

The starting balance is 5,000. The user enters menu choice 1, 2, or 3 and a non-negative amount when needed.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:
1. Check Balance
2. Deposit
3. Withdraw
Enter choice: 2
Enter deposit amount: 12000000
Balance = 12005000
>>
```

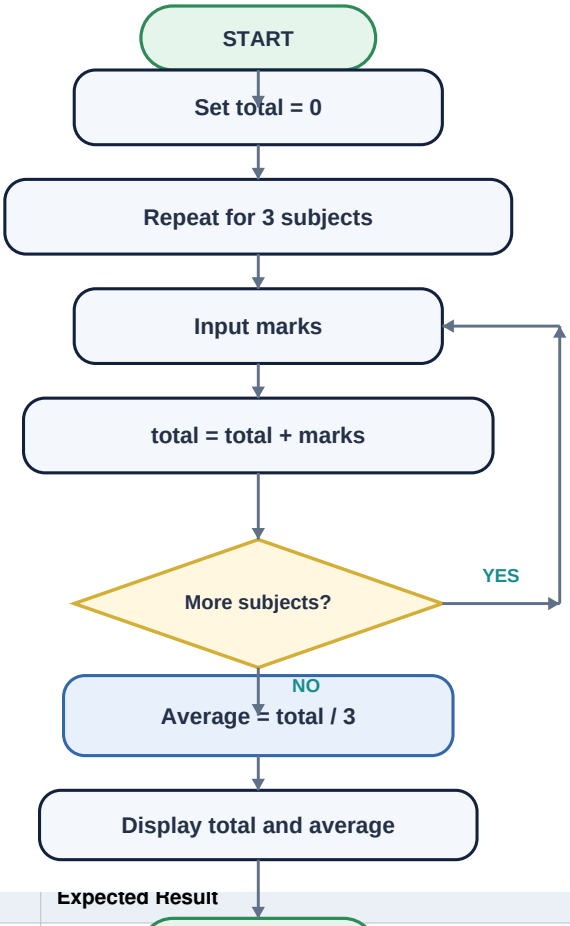
Student Marks Total and Average

Python file: a8.py **Concept:** for loop

Problem Statement

EXPERIMENT 8 | FLOWCHART

Student Marks Total and Average



Input / Test	Expected Result
80, 90, 70	Total ma = 80.0
60, 70, 80	Total marks = 210; Average marks = 70.0

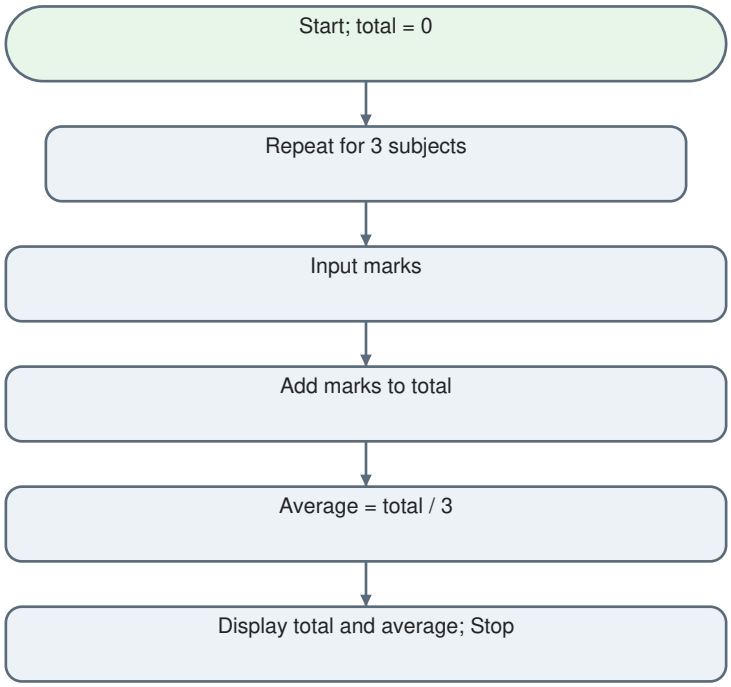
Conclusion

This case demonstrates the stated Python concept.

Experiment 8: Student Marks Total and Average

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART



ASSUMPTIONS

The user enters marks for exactly three subjects. Marks are numeric values.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\App
Enter marks: 765
Enter marks: 654
Enter marks: 765
Total marks = 2184
Average marks = 728.0
>>>
```

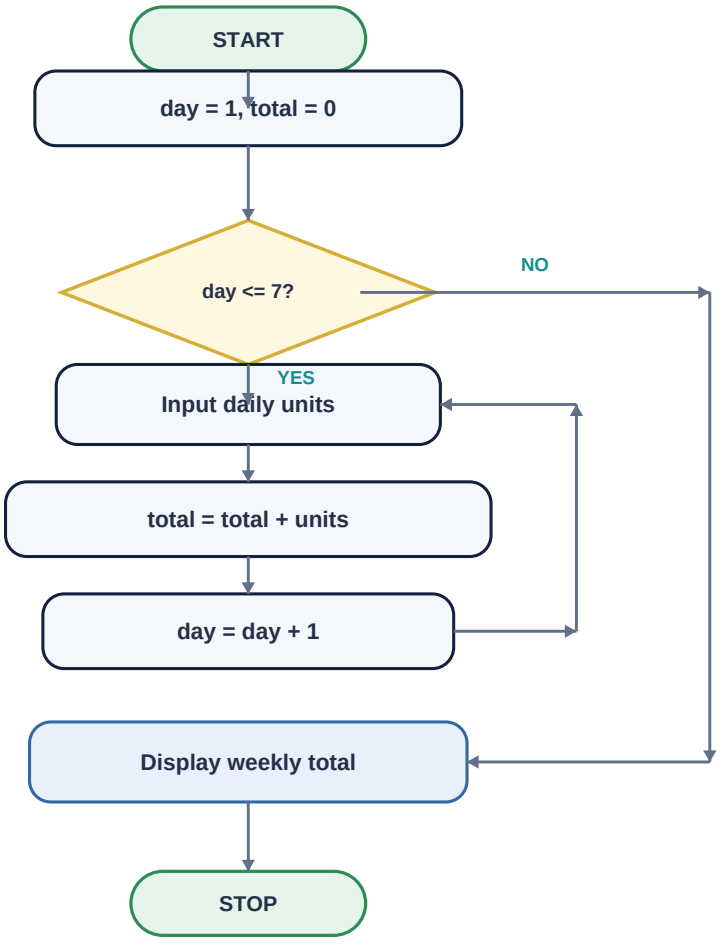
Weekly Power Consumption Tracker

Python file: a9.py **Concept:** while loop

Problem Statement

EXPERIMENT 9 | FLOWCHART

Weekly Power Consumption Tracker



Test Cases

Input / Test	Expected Result
10, 12, 8, 15, 10, 11, 14	Total weekly units = 80

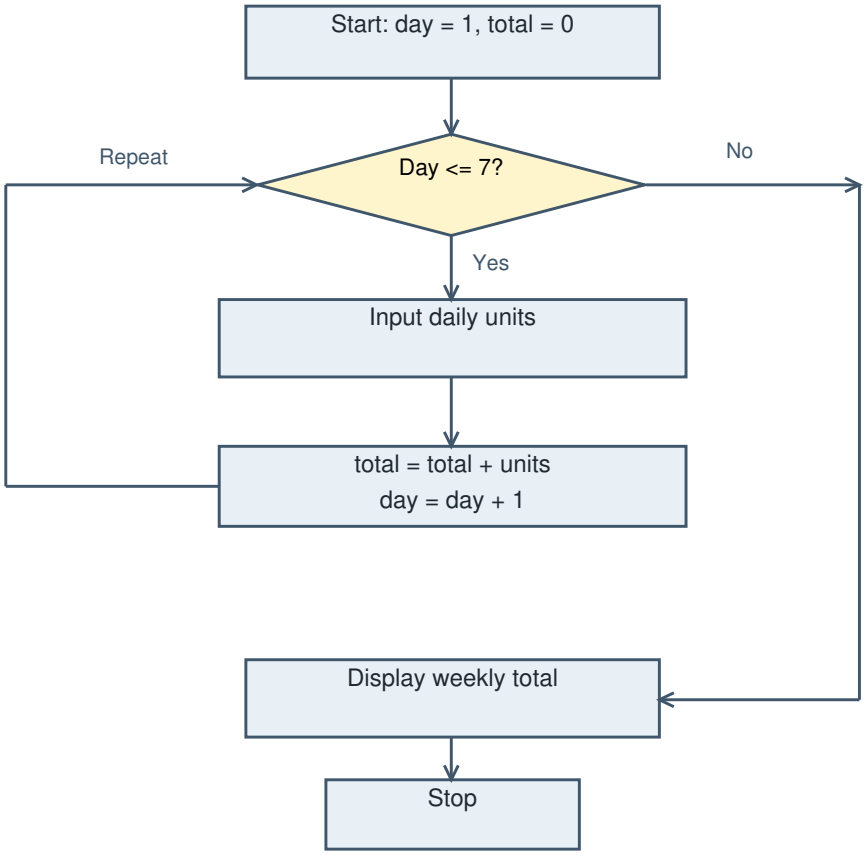
Conclusion

This case demonstrates the stated Python concept.

Experiment 9: Weekly Power Consumption Tracker

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART



ASSUMPTIONS

The user enters one non-negative unit value for each of seven days.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\AppData\Local\Programs
Enter units used: 76
Enter units used: 87
Enter units used: 87
Enter units used: 78
Enter units used: 87
Enter units used: 89
Enter units used: 98
Total weekly units = 602
>>
```

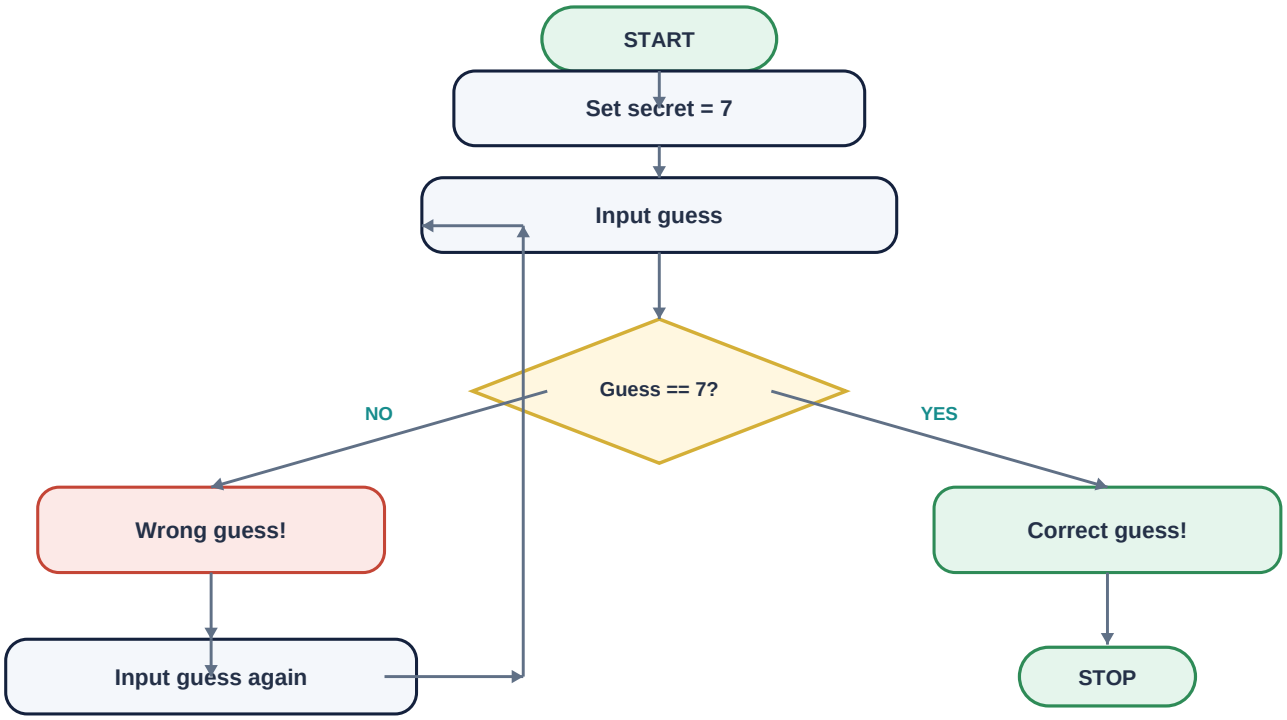
Interactive Number Guessing Game

Python file: a10.py **Concept:** while loop

Problem Statement

EXPERIMENT 10 | FLOWCHART

Interactive Number Guessing Game



Input / Test	Expected Result
4, 6, 7	Two Wrong guess! messages, then Correct guess!
7	Correct guess!

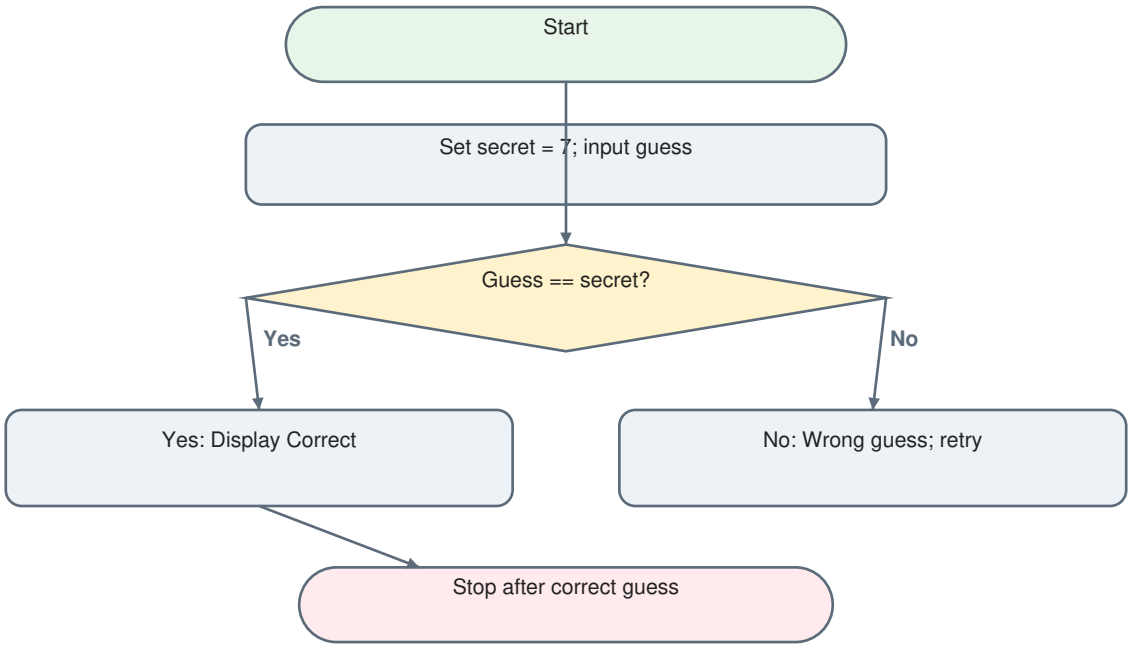
Conclusion

This case demonstrates the stated Python concept.

Experiment 10: Interactive Number Guessing Game

FLOWCHART AND OUTPUT SCREENSHOT

FLOWCHART



ASSUMPTIONS

The secret number is fixed at 7. The user enters integer guesses from 1 to 10 until correct.

PYTHON IDLE OUTPUT SCREENSHOT

```
========== RESTART: C:\Users\ASUS\AppData\Local\Programs\
Guess a number from 1 to 10: 7
Correct guess!
>
```